

$$A = \begin{bmatrix} -6 & 12 \\ -1 & 1 \end{bmatrix}, \quad P = \begin{bmatrix} -3 & -4 \\ -1 & -1 \end{bmatrix}$$

a.) Verify  $A$  is diagonalizable:

1.) Compute  $P^{-1}$ :  $\det(P) = (-3)(-1) - (-1)(-4) = -1 \neq 0$

$$P^{-1} = \frac{1}{\det(P)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{-1} \begin{bmatrix} -1 & 4 \\ 1 & -3 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ -1 & 3 \end{bmatrix}$$

2.) Compute  $P^{-1}AP$ :  $P^{-1}PA = \begin{bmatrix} 1 & -4 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} -6 & 12 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -3 & -4 \\ -1 & -1 \end{bmatrix}$

$$= \begin{bmatrix} 1 & -4 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 6 & 12 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -3 \end{bmatrix}$$

b.) Because  $A$  is similar to the diagonal matrix  $D = P^{-1}AP$ , then  $A$  and  $D$  have the same eigenvalues, and the eigenvalues are the diagonal entries of  $D$ .

Thus,  $\lambda_1 = -3$  and  $\lambda_2 = -2$        $(\lambda_1, \lambda_2) = (-3, -2)$